

Individual Project Instructions: Big Data Analytics using Hadoop (Continuous Evaluation)

Course: Big Data Analytics / Hadoop Tools

Course_Leader: Prof. Sujatha Ayyangar

Project Type: Individual Project

Project Start Date: 27.01.2026

Project Completion Date: 13.03.2026

1. Project Objective

The objective of this individual project is to help students gain **hands-on, practical experience** in working with Big Data using the Hadoop ecosystem. Students will apply concepts learned in class to a real-world dataset by performing data ingestion, preprocessing, storage in HDFS, and analytics using Hadoop tools.

By the end of this project, students should be able to:

- Understand how large datasets are handled using Hadoop.
- Preprocess real-world data for big data analytics.
- Upload and manage datasets in HDFS.
- Apply Hadoop tools for data processing and analysis.
- Interpret results and communicate insights effectively.

2. Project Domain and Dataset Selection

Each student must choose **any ONE domain**:

- Education
- Social Media
- Healthcare
- (Optional: Finance, ECommerce, Environment, Smart Cities, etc.)

Dataset Guidelines

- Download the dataset from **Kaggle or any other open data repository** (UCI ML Repository, data.gov, WHO, etc.).
- Dataset size should be **sufficiently large** to justify the use of Hadoop (minimum: 100 MB preferred after preprocessing).
- Dataset must be in **CSV / JSON / TXT / Parquet** format.
- The dataset should contain **multiple attributes** and at least one **business / social / technical problem** to analyze.

3. Project Workflow (Mandatory Steps)

Students must strictly follow the workflow below:

Step 1: Problem Definition & Domain Understanding

- Describe the chosen domain and dataset.
- Define the problem statement (e.g., “Analyzing student performance factors”, “Detecting health risk patterns”, “Understanding social media engagement trends”).
- Identify objectives and expected outcomes.

Step 2: Data Preprocessing

- Handle missing values, duplicates, and inconsistent data.
- Perform data cleaning and transformation.
- Feature selection / reduction if required.
- Save the cleaned dataset for Hadoop processing.

Step 3: Upload Dataset into HDFS

- Create directories in HDFS.
- Upload the preprocessed dataset into HDFS.
- Verify successful upload using Hadoop commands.

Step 4: Hadoop Tool Implementation (Use All Tools Taught in Class)

Students must use **all applicable tools** learned in class, such as:

- HDFS commands
- MapReduce
- Hive
- Pig
- HBase (if covered)
- Sqoop (if applicable)
- YARN (job monitoring)

Step 5: Data Analysis & Insights

- Perform analytical queries using Hive/Pig/MapReduce.
- Generate meaningful outputs (tables, summaries, trends, patterns).
- Interpret results and explain insights in the context of the chosen domain.

Step 6: Result Interpretation & Conclusion

- Discuss findings and business / social impact.

- Explain how Hadoop helped in solving the problem.
- Mention limitations and future scope.